# Assignment - 1

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# Question 1

What is a variable in C/C++?
Ans: variable is a named storage location in the computer's memory used to hold data that a program can manipulate

# Question 2

- Why do we use variables in programs?
- Ans: variables are used as named containers to temporarily store, manipulate, and retrieve data within a computer's memory during a program's execution

# Question 3

- What is the difference between a variable name and its value?
- Ans: The variable name is the identifier, while the value is the actual data stored in it.

# Question 4

- Which of the following is a valid variable name? a) 2num b) total_marks c) for d) my marks
- Ans:  b) total_marks

# Question 5

- Why is Age different from age?
- Ans:  C++ is case-sensitive, so Age and age are different variables

# Question 6

- Can a variable name contain spaces? Why?
- Ans:  No. Variable names cannot contain spaces.

# Question 7

- Can a variable name start with an underscore (_) ?
- Ans: Yes , variable names can start with an underscore . example: _count

# Question 8

- What happens if two variables have the same name in the same scope?
- Ans:  The compiler generates an error because variable names must be unique within the same
- scope

# Question 9

- Write three meaningful variable names for storing student information.
- ANS : studentName, studentAge, studentMarks

# Question 10

- Which naming convention is better and why: studentMarks or sm?
- Ans: studentMarks because it is descriptive and easy to understand

# Question 11

- What is a data type?
- Ans A data type specifies the kind of data a variable can store.

# Question 12

- Which data type is used to store whole numbers?
- Ans :  int

# Question 13

- Which data type is used to store decimal values?
- Ans: float or double

# Question 14

- What is the difference between float and double?
- Ans:  double provides greater precision than float.

# Question 15

- Which data type stores a single character?
- Ans: char

# Question 16

- Which data type stores True/False values?
- Ans:  bool

# Question 17

- What data type is suitable for storing a person's name?
- Ans: string

# Question 18

- Predict the output: int age=18; cout<<age;
- Ans: 18

# Question 19

- Identify the data type: 25, 3.14, 'A', true.
- Ans: 25=int, 3.14=double, 'A'=char, true=bool

# Question 20

- What happens if you store a decimal number in an int variable?
- Ans: The decimal part is discarded. Example: 3.14 becomes 3

# Question 21

- What is the purpose of cin?
- Ans: cin is used to take input from the user

# Question 22

- What is the purpose of cout?
- Ans:  cout is used to display output on the screen.

# Question 23

- Which symbol is used with cin?
- Ans: >>

# Question 24

- Which symbol is used with cout?
- Ans: <<

# Question 25

- Explain: cin >> age;
- Ans:It takes input from the user and stores it in age.

# Question 26

- Explain: cout << age;
- Ans: It displays the value of age

# Question 27

- What is the purpose of endl?
- Ans:  endl moves the cursor to the next line.

# Question 28

- Write a statement to input a student's marks.
- Ans: cin >> marks;

# Question 29

- Write a statement to display 'Welcome to C++’.
- Ans: cout << "Welcome to C++";

# Question 30

- Predict the output: cout << 'Programming’;
- Ans: Error because single quotes are for a single character.

# Question 31

- What is an operator?
- Ans: An operator is a symbol that performs an operation on operands

# Question 32

- Which operator is used for addition?
- Ans:+

# Question 33

- What is the result of 10 % 3?
- Ans: 1
  Explanation: The %(modulus)operator gives the remainder after division.
  10 % 3 = 1
  Because: 10/3=3 remainder 1.

# Question 34

- What is the result of 8 % 2?
- Ans:0
- Because:8 is completely divisible by 2,so remainder is 0.

# Question 35

- Which operator checks equality?
- Ans:==

# Question 36

- What is the difference between = and == ?
- Ans:= assigns a value, while == compares two values.

# Question 37

- What is the result of 5 > 3?
- Ans:True

# Question 38

- What does && represent?
- Ans: Logical AND operator.It returns true only when both conditions are true.

# Question 39

- What does || represent?
- Ans: Logical OR operator.It returns true if at least one condition is true .

# Question 40

- What does ! represent?
- Ans: Logical NOT operator.It reverses the result of a condition.

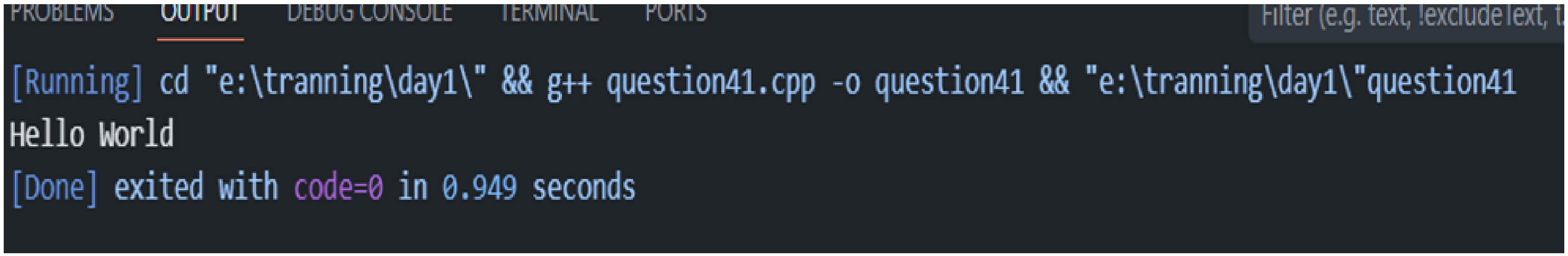
# Question 41

- Write a program to print”Hello world”?
- **Problem Statement:**
- Write a program to display”Hello world “ on screen.
- **Algorithm:** 1. start the program.
  2.Display “Hello world” using cout3.
  3.End the program.
- **C++ Program**

```
#include<iostream>
using namespace std;
int main()
{
cout<<“Hello world”;
return 0;
}
```

- **Explanation :**
- #include <iostream>: used for input/output operations
- using namespace std; → Allows use of cout without std::

- cout → Displays output on screen
- return 0; → Ends the program successfully
- **Output**



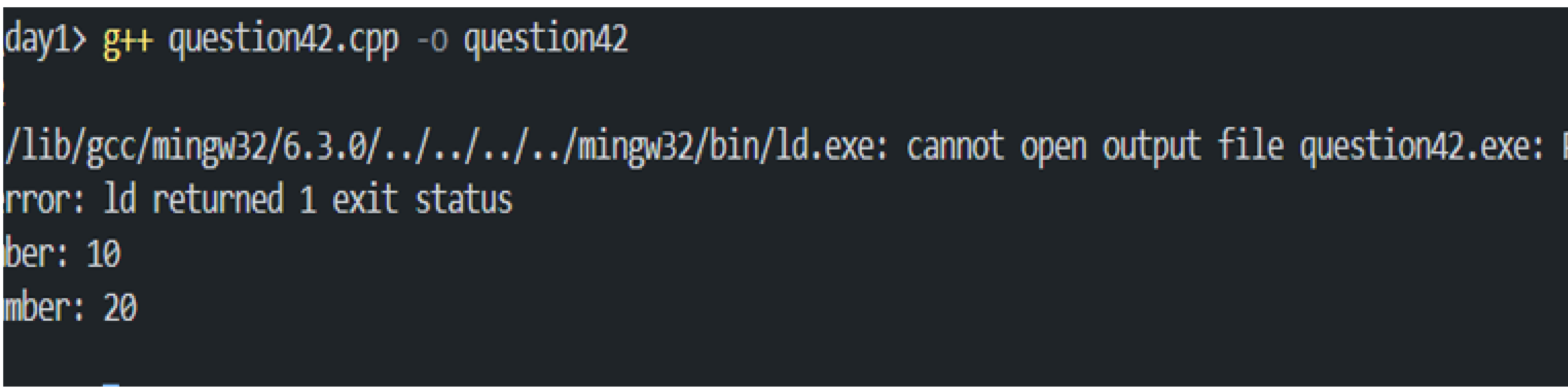
# Question 42

- Write a program to input two numbers and display their sum.
- **Problem Statement**
- Write a program to take two numbers from the user and display their sum.
- **Algorithm**
    1. Start the program
    2. Declare two variables
    3. Take input from user
    4. Add both numbers
    5. Display the result
    6. End the program
- **C++ Program**

```
#include<iostream>
Using namespace std;
int main()
{
int num1,num2,sum;
```

```
cout<<“enter first number:”;
cin>>num1;
sum=num1 +num2;
cout<<“sum=“<<sum;
Return 0;
}
```

**Explanation**



# Question 43

- Write a program to calculate the area of a rectangle.
- **Problem Statement**
  Write a program to calculate the area of a rectangle.
- **Formula:**
  Area = Length × Breadth
- **Algorithm**
  1. Start the program
  2. Declare variables for length and breadth
  3. Take input from user
  4. Calculate area
  5. Display area

1. End the program

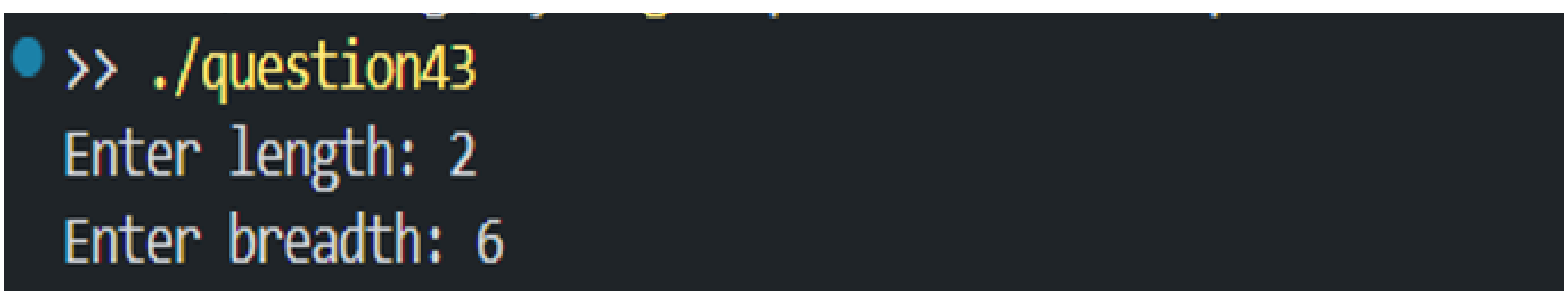
**C Pgrogram**

```
#include<stdio.h>
int main()
{
```

```
float length, breadth, area;
    printf("Enter length: ");
    scanf("%f", &length);
   printf("Enter breadth: ");
    scanf("%f", &breadth);
  area = length * breadth;
  printf("Area of Rectangle = %.2f", area);
 return 0;
}
```

**Explanation**

- scanf() takes input
- length * breadth calculates area
- printf() displays output
- %.2f shows decimal values up to 2 places



# Question 44

- What formula is used to calculate Simple Interest?
- **Ans:**The formula used to calculate Simple Interest (SI) is:
  SI = P*R*T/100
  Where:
  - P = Principal Amount
  - R = Rate of Interest
  - T = Time
  - SI = Simple Interest

# Write a program to calculate Simple Interest.

- **Problem Statement**
  Write a program to calculate Simple Interest using principal, rate, and time.
- **Algorithm**
  1. Start the program
  2. Declare variables
  3. Take principal, rate, and time as input
  4. Apply formula
  5. Display result

1. End the program

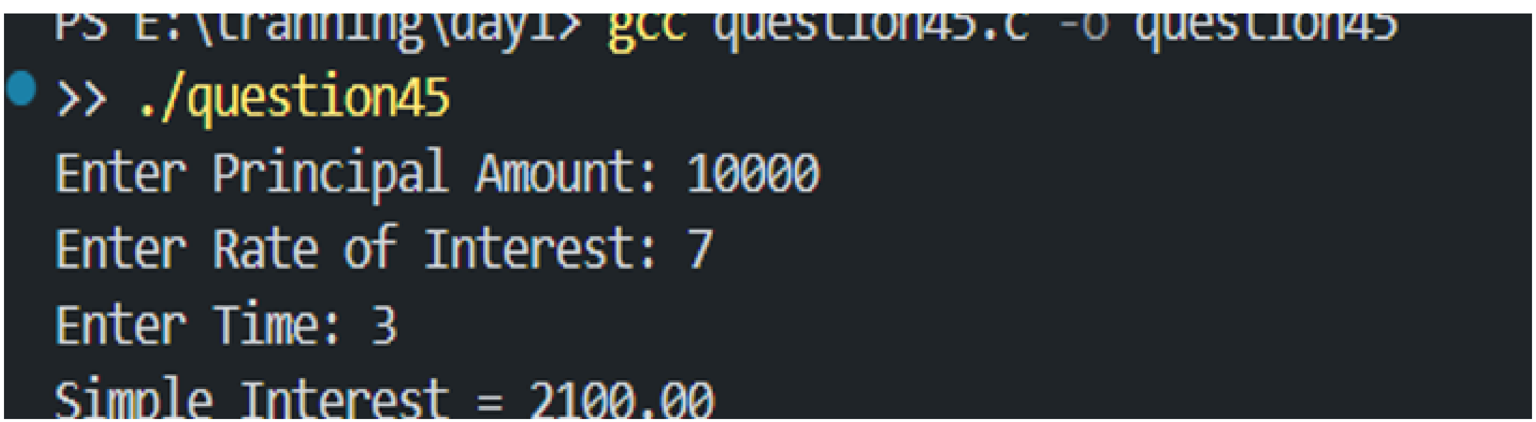
**C Program**

```
#include<stdio.h>
int main()
{
Float principal,rate,time,SI;
printf("enter principal amont:");
scanf("%f",&principal);
```

```
 printf("Enter Rate of Interest: ");
   scanf("%f", &rate);
   printf("Enter Time: ");
   scanf("%f", &time);
   SI = (principal * rate * time) / 100;
   printf("Simple Interest = %.2f", SI);
   return 0;
}
```

**Explanation**

- User enters principal amount, rate, and time
- Formula is applied to calculate simple interest
- Result is displayed using printf()



```
PS E:\tranning\day1> gcc question45.c -o question45
>> ./question45
Enter Principal Amount: 10000
Enter Rate of Interest: 7
Enter Time: 3
Simple Interest = 2100.00
```

# Question 46

- Why is a temporary variable used while swapping two numbers?
- **Ans** : A temporary variable is used to safely store one value while swapping two numbers so that the original value is not lost.

# Question 47
# Swap the values: a=10, b=20

- **Problem Statement**
  Write a program to swap two numbers using a temporary variable.
- **Algorithm**
  1. Start the program
  2. Declare variables a, b, and temp
  3. Store value of a in temp
  4. Assign value of b to a
  5. Assign value of temp to b
  6. Display swapped values

1. End the program

**C++ Program**

```
#include<iostream.h>
Using namespace std;
int main()
{
int a=10,b=20,temp;
```

```
cout << "Before Swapping:" << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;
   temp = a;
    a = b;
    b = temp;
    cout << "After Swapping:" << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;
    return 0;
}
```

**Explanation**

- temp = a stores value of a
- a = b copies value of b into a
- b = temp restores original value of a into b
- This swaps both numbers successfully

```
>> ./question47
Before Swapping:
a = 10
b = 20
After Swapping:
a = 20
```

# Question 48

- What formula is used to convert Celsius into Fahrenheit?
- **Ans:** $F=9/5C+32$
- **Explanation:**
  C= temperature in Celsius
  F= temperature in fahrenheit

# Question 49

- What is the Fahrenheit value of 0°C?
- **Ans:** using the formula F=9/5C+32

  Where C = 0°c
- **Calculation:**

  $F = (9/5 \times 0) + 32$

  $F = 32°F$
- **Final Answer:**

  The Fahrenheit value of 0°C is 32°F.

# Why can integer division give incorrect results in the Celsius-to-Fahrenheit program?

## Answer

Integer division can give incorrect results because integers remove the decimal part.

**For example**:

5 / 9

This becomes:

0

instead of:

0.555...

So, if integer division is used in the Celsius-to-Fahrenheit formula, the answer may become wrong.

**Correct Way**

Use float or double for accurate decimal calculations.